

Sree Nayanika Kadudas

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CERTIFICATIONS: AWS Certified Developer – Associate

Profile

Senior Software Engineer with 10 years of experience building AI driven applications and cloud native backend systems. Strong expertise in Python development, Generative AI, and RAG architecture using LangChain, FAISS, Pinecone, and Hugging Face. Experienced in developing scalable FastAPI/Flask microservices and deploying high availability solutions across AWS and Azure. Skilled in Docker, Kubernetes, CI/CD pipelines, and observability for distributed systems, with a strong focus on building production grade GenAI platforms and mentoring engineering teams.

Technical skills

Category	Technologies / Tools
Generative AI & LLMs	Large Language Models, Generative AI, RAG Architectures, Prompt Engineering, Multi-Agent Workflows, NLP, Deep Learning
Programming Languages	Python, SQL, JavaScript, TypeScript, Bash / Shell Scripting
AI / ML Libraries	PyTorch, TensorFlow, Hugging Face Transformers
Gen AI Tooling	LangChain, OpenAI GPT-4, Anthropic Claude, AWS Bedrock
Data Processing	ETL Workflows, Pandas, NumPy, Batch Processing, Asynchronous Data Pipelines
Backend Development	REST APIs, FastAPI, Flask, Django, Microservices Architecture
Databases and Search	MySQL, PostgreSQL, MongoDB, DynamoDB, Redis, Elasticsearch
AWS Platform	AWS Lambda, Step Functions, Glue, Athena, EC2, S3, RDS, API Gateway, SQS, EventBridge
Containers & Infrastructure	Docker, Kubernetes (AWS EKS), Terraform
CI/CD	Jenkins, GitLab CI/CD, GitHub Actions, AWS CodePipeline
MLOps	MLflow, SageMaker Processing / Batch Inference, Model Monitoring
Security	AWS IAM, AWS KMS, AWS Secrets Manager, OAuth2, JWT, VPC

Monitoring & Logging	Prometheus, Grafana, CloudWatch, ELK Stack, AWS X-Ray
Version Control	Git, GitHub, Bitbucket

Professional Experience

Software Engineer

March 2025 - Present

Honeywell International Inc.

Design and build scalable Python based services and AI driven applications supporting cloud native platforms and enterprise automation. Develop microservice architectures and integrate Generative AI capabilities to enhance data driven workflows, operational intelligence, and system performance across distributed environments.

- Develop and maintain **Python/FastAPI microservices** supporting scalable backend systems and enterprise AI applications. Integrate **Generative AI and conversational AI capabilities** using OpenAI, Anthropic, Hugging Face, and AWS Bedrock to enable enterprise AI applications with controlled inference behavior.
- Build and integrate **Generative AI and RAG based solutions** using LangChain, OpenAI, Anthropic, and Hugging Face to enable intelligent search, conversational interfaces, and knowledge retrieval workflows.
- Implement **semantic retrieval pipelines** using FAISS and Pinecone with embedding models to support high accuracy document search and context aware responses.
- Design and deploy **cloud native services on AWS** using EC2, Lambda, S3, DynamoDB, RDS, and API Gateway while ensuring reliability, scalability, and security.
- Develop **asynchronous data processing pipelines** using Python, AWS Lambda, Step Functions, and Glue for automated ingestion and preprocessing of large datasets.
- Implement **event driven architectures** using SNS and SQS to enable decoupled communication between distributed microservices.
- Improve **observability and monitoring** using CloudWatch, Prometheus, Grafana, and ELK Stack to proactively identify and resolve production issues.
- Deploy and manage containerized services using **Docker and Kubernetes (EKS)** and maintain CI/CD pipelines with Jenkins, GitHub Actions, and GitLab CI for automated builds and deployments.

- Optimize system performance through **Redis caching, vector search optimization, and efficient API design** to reduce latency and improve service reliability.
- Participate in **architecture design discussions, code reviews, and incident response**, contributing to system stability and long-term platform improvements.
- Collaborate with cross functional teams including product managers, data engineers, and platform engineers to deliver scalable enterprise solutions.

Environment: Python, FastAPI, Flask, Generative AI / LLMs, RAG Architectures, LangChain, Prompt Engineering, Multi Agent Workflows, OpenAI GPT-4, Anthropic Claude, Hugging Face Transformers, Vector Embeddings, FAISS, Pinecone, Redis, Document Processing (PDF/ETL), AWS Lambda, AWS Step Functions, AWS Glue, AWS EC2, AWS S3, AWS RDS, AWS DynamoDB, API Gateway, SNS, SQS, Docker, Kubernetes (AWS EKS), CI/CD (Jenkins, GitHub Actions, GitLab CI), Prometheus, Grafana, ELK Stack, Amazon CloudWatch, Git, Linux, REST APIs, OAuth2, JWT, AWS IAM, AWS KMS, AWS Secrets Manager, VPC.

Software Engineer

September 2023 – February 2025

Morgan Stanley, US

Developed scalable Python based backend services and AI driven solutions supporting enterprise applications and data driven platforms. Built microservice architecture and integrated machine learning and generative AI capabilities to improve system intelligence, automation, and operational efficiency across distributed cloud environments.

- Develop and maintain **Python/FastAPI microservices** supporting scalable backend platforms and enterprise applications. Build secure REST APIs to enable reliable communication between distributed systems.
- Implement **Generative AI and conversational AI integrations** using OpenAI, Anthropic, and Hugging Face models to support intelligent automation and knowledge retrieval workflows.
- Design and implement **retrieval pipelines and semantic search capabilities** using FAISS and Pinecone with embedding models to improve document search accuracy and contextual response generation.
- Develop **data processing and ETL pipelines** using Python, Pandas, and SQL to ingest, transform, and analyze large datasets supporting analytics and machine learning workflows.
- W Deploy and manage **cloud native services on AWS** including EC2, Lambda, S3, DynamoDB, and RDS while ensuring system reliability, scalability, and security.

- Build **event driven architectures** using SNS and SQS to enable asynchronous communication between microservices and distributed applications.
- Improve system monitoring and observability using Amazon CloudWatch, Prometheus, Grafana, and ELK Stack to track performance metrics and detect operational issues.
- Containerize services using **Docker** and deploy applications on **Kubernetes (EKS)** while implementing CI/CD pipelines with Jenkins and GitHub Actions for automated builds and deployments.
- Collaborate with product managers, data engineers, and platform teams to deliver secure and scalable enterprise applications aligned with organizational architecture standards.

Environment: Python, FastAPI, Generative AI / LLMs, LangChain, OpenAI, Hugging Face Transformers, FAISS, Pinecone, Pandas, SQL, AWS (EC2, Lambda, S3, RDS, DynamoDB, API Gateway, SNS, SQS), Docker, Kubernetes (EKS), Jenkins, GitHub Actions, Prometheus, Grafana, ELK Stack, Amazon CloudWatch, Git, Linux, REST APIs, OAuth2, JWT, IAM, VPC.

Software Engineer

March 2022 – July 2023

Databricks

Developed and maintained scalable backend platforms supporting high traffic web applications and distributed services. Designed microservice based systems and cloud native applications while improving system reliability, deployment automation, and performance across cloud infrastructure.

- Design and develop **Python based backend services** supporting scalable web applications and internal business platforms.
- Build and maintain **RESTful APIs** to enable secure communication between frontend systems, backend services, and external integrations.
- Architect and maintain **microservice based application components** to support modular system design and independent service deployments.
- Design and optimize **database schemas and complex SQL queries** to improve application performance and data retrieval efficiency.
- Deploy and manage applications on **AWS infrastructure** including EC2, S3, and RDS while improving system scalability and uptime.
- Implement **containerized application environments using Docker** to streamline development and deployment workflows.
- Develop **automated deployment pipelines** using Jenkins and GitHub Actions to support continuous integration and faster release cycles.

- Improve **application performance and reliability** by identifying bottlenecks, optimizing backend logic, and implementing caching strategies.
- Monitor application health and system performance using **CloudWatch, Prometheus, and Grafana** to proactively identify operational issues.
- Collaborate with engineering, product, and operations teams to deliver new platform features and maintain stable production environments.

Tech Stack: Python, Django, Flask, FastAPI, JavaScript, HTML5, CSS3, React.js, REST APIs, Microservices Architecture, SQL, MySQL, PostgreSQL, MongoDB, Git, Agile/Scrum, JIRA, Docker, Jenkins, GitHub Actions, AWS (EC2, S3, RDS), Linux, Unix, Bash, Pandas, NumPy, JSON.

Software Developer

October 2019 - February 2022

Siemens

Developed scalable web applications and backend systems supporting enterprise platforms and customer-facing applications. Worked across full stack development, API design, and cloud deployments while improving application performance, data processing workflows, and system reliability in production environments.

- Develop and maintain **Python/Django backend services** supporting web applications and business platforms.
- Design and implement **RESTful APIs** to enable secure integration between frontend systems, backend services, and third-party applications.
- Build responsive **user interfaces using JavaScript, HTML5, CSS3, and React.js** to improve usability and performance of web applications.
- Design and optimize **relational databases and SQL queries** using MySQL and PostgreSQL to support high volume transactional systems.
- Deploy and maintain applications on **cloud infrastructure including AWS EC2, S3, and RDS**, ensuring reliability and scalability of services.
- Containerize applications using **Docker** and support deployment automation through **Jenkins and Git-based workflows**.
- Monitor application performance and troubleshooting production issues using **Splunk and CloudWatch** to maintain system stability.
- Collaborate with cross functional teams including product managers, QA engineers, and front-end developers to deliver new platform features.
- Participate in **Agile development processes, sprint planning, and code reviews** to maintain development standards and improve delivery timelines.

Environment: Python, Django, JavaScript, React.js, HTML5, CSS3, REST APIs, SQL, MySQL, PostgreSQL, Pandas, AWS (EC2, S3, RDS), Docker, Jenkins, Git, JIRA, Splunk, Linux.

Software Developer

May 2016 - September 2019

Tata Consultancy Services

Contributed to the development and enhancement of enterprise web applications used by internal business teams and external clients. Focused on backend development, database integration, and application maintenance while supporting large-scale systems in a structured enterprise development environment.

- Implement backend application components using **Python and Django** to support enterprise business workflows and internal tools.
- Create and maintain **API endpoints** enabling integration between application modules and external service platforms.
- Develop interactive **frontend components using JavaScript, HTML5, and CSS3** to improve application functionality and usability.
- Write and optimize **SQL queries and stored procedures** to support data processing and reporting requirements.
- Perform **application debugging and issue resolution** to address production defects and improve system reliability.
- Assist in implementing **data validation, error handling, and logging mechanisms** to enhance application stability.
- Support **application deployment and configuration in Linux environments**, ensuring proper system setup and operational readiness.
- Work closely with QA teams during **system testing and release cycles**, resolving defects and improving software quality.
- Participate in **Agile sprint cycles, requirement analysis, and technical discussions** to support timely feature delivery.
- Maintain project repositories and collaborate with teams using **Git-based version control workflows**.

Education

Master's Business Analytics
University of North Texas, USA

Bachelor of Technology
Osmania University, India